



Unit 4 · Outcome 2 · Energy Sources

Keeping the lights on: base and peak load

KK09

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



01 Electricity can't be stockpiled — supply must equal demand every single second

Coal, gas, water and sunshine can be stored. The electricity made from them, mostly, cannot. The amount a grid generates has to match the amount everyone is using at that instant, or the system's frequency drifts away from 50 Hz and protective relays start tripping — that's a blackout.

◆ ANALOGY

Analogy — the tank with no tank Picture a water tower that supplies a town but has no actual storage tank — water flows straight from the pumps to every tap. If everyone showers at 7 am and the pumps don't speed up, pressure collapses. If the pumps over-pump, pipes burst. The electricity grid runs exactly like this: generation in must equal consumption out, moment to moment. "Storage" (batteries, pumped hydro) is the expensive, recently-added exception — and it's the whole reason base/peak load is a live debate.

The 24-hour shape of that demand is the key to the entire dot point. Demand is never flat: it has a floor that is always there, and spikes at predictable times. Engineers split the curve into two ideas — base load and peak load — because the cheapest, cleanest way to meet each is completely different.

BASE VS PEAK LOAD

02 Reading the demand curve

Over any 24 hours, total electricity demand rises and falls in a familiar pattern: low overnight, a morning lift, a dip through the working day, then a sharp evening peak as people get home, cook, heat or cool, and switch everything on at once.

The minimum, continuous level of demand that is present at every hour of the day — the floor of the curve. It comes from things that never switch off: refrigeration, hospitals, water and sewage pumps, street lighting, data centres, and industry running through the night.

On the Yarrabin grid the overnight minimum sits near 2,300 MW — that is the base load.

The maximum demand, reached only briefly — the crest of the curve. It's driven by everyone using power at the same time: the ~6–8 pm cooking/heating/cooling surge, or a 40°C summer afternoon when air-conditioners run flat out.

Yarrabin's evening peak hits about 4,900 MW — more than double the base.

◆ ANALOGY

Analogy — your phone's battery drain Base load is the background drain — the clock, the radios, the always-syncing apps — a small, steady draw you can't switch off. Peak load is opening a graphics-heavy game — a short, intense spike on top of the background. You wouldn't engineer the whole phone around the game, but you do need headroom to handle it when it happens.

◆ KEY INSIGHT

The examinable distinction Base load is not "the biggest source of energy" and peak load is not "renewable energy." They are parts of the demand curve: base = the always-present floor; peak = the brief maximum. Sources are then judged by how well they can supply each.

MATCHING SOURCES TO LOAD

03 Dispatchability: can you turn it up when you need it?

The single property that decides whether a source suits base or peak load is dispatchability — whether grid operators can raise or lower its output on command. This splits every source into three groups.

Role	Behaviour	Best-suited sources	Why
Base-load	Runs continuously at a steady output; slow to start up or shut down	Black/brown coal, nuclear, large hydro, geothermal, biomass	Cheap to keep running once hot; high capacity factor (~80–90%); inefficient and costly to keep ramping
Peaking	Dispatchable — starts, ramps and stops quickly	Open-cycle gas turbines, hydro (esp. pumped hydro), grid batteries	Fast response fills short spikes; higher cost per unit, so only run during peaks
Variable (intermittent)	Output set by weather/time, not by the operator — must-take	Solar (midday only), wind (when it blows)	Zero fuel cost & emissions, but can't be “turned up” for the evening peak; needs firming

✓ TIP

Capacity ≠ energy Capacity (MW) is how much a generator can produce at full tilt. Energy (MWh) is how much it actually delivers over time. A wind farm rated 100 MW with a 35% capacity factor delivers, on average, only 35 MW of energy — and you can't choose when. Examiners love this gap: high capacity does not mean reliable supply.

GRID DISPATCH SIMULATOR

04 Run the Yarrabin grid for a day

This is the dot point made tangible. The black line is the real demand curve from §02. Your job: build a supply stack beneath it that meets demand at every hour — without big shortfalls (blackouts) or huge surpluses (wasted, curtailed power). Adjust each generator and watch the stack and the gauges respond.

✓ TIP

Try this sequence 1) Hit “coal only” — reliable, but watch the night-time surplus (wasted) and the sluggish evening crest. 2) Hit “solar + wind only” — clean, but the supply collapses at the 6 pm peak: a blackout. 3) Hit “balanced firm mix” — coal/biomass holds the floor, solar/wind do the daytime, and battery + hydro + gas firm the evening peak. That last move is the modern energy transition in one chart.

THE DUCK CURVE

05 Why more solar makes the evening peak harder

Here's the twist that makes this dot point genuinely current. Rooftop and farm solar don't reduce demand evenly — they slash it only in the middle of the day. What the grid's big generators actually have to supply is net demand = total demand – solar. As solar grows, that net-demand curve develops a deep midday belly and a steep evening neck. Plotted out, it looks like a duck.

⚠ WATCH OUT

The duck's neck is the problem. The issue isn't midday — there's often too much solar then (prices go negative; output is curtailed). The issue is the evening ramp: as the sun sets, thousands of megawatts must come online in a couple of hours, right as demand peaks. Only fast, dispatchable supply — pumped hydro, batteries, gas — can climb that neck. This is precisely why “just build more solar” doesn't finish the job.

◆ ANALOGY

Analogy — the all-you-can-eat lunch. Solar is a generous friend who shouts everyone a huge free lunch at midday — but vanishes by dinner. Lunch is so big some food is wasted (curtailment). Then at 7 pm everyone's hungry again and your friend is gone, so you have to cook a full meal, fast. The bigger the free lunch, the more dramatic the dinner scramble. Storage is a fridge: it lets you keep some of that free lunch for dinner.

CURRENT & PROJECTED NEEDS

06 From “current” to “projected” — what the VCAA wording means

The dot point asks about current and projected needs. These describe two different grids.

- Base load still leans on ageing coal (e.g. Loy Yang in the Latrobe Valley) and large hydro.
- Peaks met by gas turbines and hydro.
- A fast-growing slice of variable solar and wind, plus the world's highest rooftop-solar uptake.
- Coal stations retire — the base-load floor must be held a new way.
- Far more variable renewables → a deeper duck → more need for firming.
- Storage scales up: pumped hydro (Snowy 2.0), big batteries (Victorian Big Battery, Geelong), home batteries, possibly hydrogen.
- Interconnectors share supply between states; demand management shaves peaks.

◆ KEY INSIGHT

The “firming” idea — write this in evaluation answers. Variable renewables can supply energy cheaply but cannot, alone, guarantee base or peak load on demand. Firming means pairing them with storage and/or fast dispatchable backup so the combined system is reliable. “Renewables + storage” can meet base and peak load; “renewables alone” cannot. That distinction earns marks.

The real Australian context (grounding the Yarrabin fiction): the National Electricity Market (NEM) connects the eastern states; the Australian Energy Market Operator (AEMO) forecasts demand and dispatches generators every 5 minutes to keep supply matched to load. Its Integrated System Plan projects coal exiting and storage + transmission filling the gap.

INDIVIDUAL & SOCIETAL

07 How base and peak needs are met — at two levels

The dot point ends with “at individual and societal levels.” Meeting demand isn't only an engineering job for big operators; households shape the curve too. A strong answer addresses both.

- Generation mix — build base-load + peaking + variable + storage so the stack can meet any hour.
- Dispatch & markets — AEMO/the NEM forecast demand and dispatch the cheapest available generators in real time.

- Storage & firming — pumped hydro and grid batteries store midday surplus for the evening peak.
- Interconnectors — move power between regions to cover local shortfalls.
- Demand response — pay big users to cut load during extreme peaks.
- Rooftop solar + home battery — self-supply by day, store for the evening peak.
- Load shifting — run dishwashers, pool pumps and EV charging off-peak (overnight), flattening the curve.
- Time-of-use tariffs — cheaper off-peak prices nudge this behaviour.
- Energy efficiency — better insulation, efficient appliances and reverse-cycle heating cut both base and peak draw.
- EVs as storage — vehicle-to-grid (V2G) can feed stored charge back during peaks.

✓ **TIP**

The link between the levels Individual actions aggregate into societal effects. A million homes shifting their dishwasher to 11 pm visibly lowers the evening peak — meaning society needs fewer expensive peaking plants. That's why utilities reward off-peak use: shaving the peak is cheaper than building plant to meet it.

COMMAND WORDS

08 Decode what the question is really asking

◆ **COMMAND WORDS**

The verb sets the marks. Tap a command word to see what an examiner expects on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

EXAM TRAPS

09 Six ways students lose marks here

◆ **NOTE**

1 · “Base load = the most energy” No. Base load is the minimum continuous demand (the floor), not the largest amount of energy. Define it as a level of demand, not a quantity of supply.

◆ **NOTE**

2 · “Solar/wind can provide base load” Alone, no — they're variable, not continuous or dispatchable. Say “solar + storage can help meet base load,” never solar by itself.

◆ NOTE

3 · Confusing capacity with energy A 100 MW wind farm rarely delivers 100 MW. Distinguish installed capacity (MW) from delivered energy (MWh) via capacity factor.

◆ NOTE

4 · “Batteries generate electricity” They don’t — they store and release it. They shift energy in time (charge midday, discharge at peak). Call them storage, not generation.

◆ NOTE

5 · Ignoring “individual and societal” If the question includes both levels, an answer about only the grid (or only households) is capped. Cover both and link them.

◆ NOTE

6 · Forgetting the instantaneous match The reason peak load matters is that supply must equal demand right now. Mention the matching/reliability constraint to show you understand why peaking sources exist.

P·E·L TRAINER

10 Write a paragraph that earns the marks



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

VCE extended responses reward structure. Use P·E·L: make a Point, support it with Evidence (a named source, figure or mechanism), then Link back to the question. Type a response below — the meters detect whether each element is present and flag key terminology.

WORKED ANSWER

11 Same question, two responses

Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don’t make as much power. So coal is more reliable but batteries are cleaner.

• NOTE

Why it loses marks: no base/peak framing; “batteries are renewable” is wrong (they’re storage, not a source); “don’t make as much power” confuses capacity with role; it’s two separate descriptions, not a true comparison; no dispatchability or capacity-factor language.

Brown coal suits base load: it runs continuously at a high capacity factor (~85%), so it economically holds the always-present demand floor — but it ramps slowly, so it cannot follow short spikes, and it emits heavily.

Grid batteries suit peak load: they are storage, not generation, dispatching stored energy within seconds to cover the brief evening peak, with no direct emissions — but their stored energy is limited, so they cannot sustain base load for long unrecharged.

Overall, the two are complementary rather than competing: coal (or a low-emission base-load source) holds the floor while batteries firm the peak — so a reliable grid needs both roles filled, not one source.

• **NOTE**

Why it scores: explicit base- vs peak-load roles; correct “storage not generation”; capacity factor and dispatchability used precisely; genuine comparison with a synthesising judgement; notes a limitation of each.

PRACTICE QUESTIONS

12 Five graded questions · 20 marks

★ **PRACTICE & SELF-MARK**

Attempt each in writing first, then reveal the marking guide and self-mark. Your score feeds the dock in the corner.

KEY TAKEAWAYS

13 If you remember nothing else

♦ **RECAP / CHEAT SHEET**

Supply must match demand every second. Electricity is barely storable, so the 24-hour demand curve drives everything. Base load = the continuous minimum demand (the floor); peak load = the brief maximum (the evening/heatwave crest). They're parts of the demand curve, not types of source. Dispatchability decides the match. Continuous, slow sources (coal, nuclear, geothermal, large hydro) suit base load; fast, dispatchable sources (gas, hydro, batteries) suit peaks. Variable renewables (solar, wind) supply cheap energy but can't be turned up on demand — so alone they meet neither base nor peak reliably. They need firming (storage + backup). The duck curve: rooftop solar guts midday net demand and steepens the evening ramp — making storage, not more solar, the key to the peak. Capacity (MW) ≠ energy (MWh). Capacity factor bridges them. Batteries store, they don't generate. Two levels: societal (generation mix, AEMO dispatch, storage, interconnectors, demand response) and individual (rooftop solar + battery, load shifting, efficiency, EVs) — and individual actions aggregate to shave the societal peak.

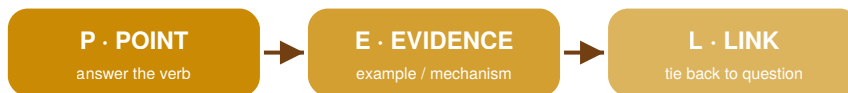
◆ **KEY INSIGHT**

One-line summary A reliable grid fills two jobs — a steady base-load floor and a fast peak-load crest — and the energy transition is really about meeting both as we swap dispatchable fossil fuels for variable renewables backed by storage, at both the grid and household level.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

1.5/4 Coal is better than batteries because it makes a lot of electricity all the time. Batteries are renewable and good for the environment but they don't make as much power. So coal is more reliable but batteries are cleaner. Why it loses marks: no base/peak framing; "batteries are renewable" is wrong (they're storage, not a source); "don't make as much power" confuses capacity with role; it's two separate descriptions, not a true comparison; no dispatchability or capacity-factor language.

✓ STRONG / MODEL ANSWER

Brown coal suits base load: it runs continuously at a high capacity factor (~85%), so it economically holds the always-present demand floor — but it ramps slowly, so it cannot follow short spikes, and it emits heavily. Grid batteries suit peak load: they are storage, not generation, dispatching stored energy within seconds to cover the brief evening peak, with no direct emissions — but their stored energy is limited, so they cannot sustain base load for long unrecharged. Overall, the two are complementary rather than competing: coal (or a low-emission base-load source) holds the floor while batteries firm the peak — so a reliable grid needs both roles filled, not one